BPS-17 Lecturer Mathematics
AJKPSC / FPSC / PPSC / KPSC
Comprehensive Study Plan

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Differential Equations — Complete MCQ Study Plan

Topics Covered:

- Formation of ODEs
- First Order & First Degree ODEs
- Separable Equations
- Homogeneous Equations
- Exact Equations & Int. Factors
- Linear First-Order ODEs
- Bernoulli Equation
- Orthogonal Trajectories
- Second-Order Linear ODEs
- Homogeneous with Const. Coefficients
- Undetermined Coefficients
- Variation of Parameters
- Cauchy–Euler Equation
- Power Series Solutions
- Laplace Transforms
- Partial Differential Equations (Intro)

Intermediate Level: **Concept Summary → Key Formulas → MCQs with Full Explanations**

Reference Material — PSC 2021–26

Contents

1. Formation of Ordinary Differential Equations

An **ordinary differential equation (ODE)** is an equation involving an unknown function $y(x)$ and its derivatives $y', y'', \ldots$. It is formed by *eliminating arbitrary constants* from a family of curves. If a family has n arbitrary constants, differentiating n times and eliminating constants yields an ODE of **order** n.

Order: highest derivative present. **Degree:** power of the highest derivative (after clearing radicals/fractions).

Key Formulas & Results

- Family with n arbitrary constants $\longrightarrow$ ODE of order n
- **General solution:** contains n arbitrary constants
- **Particular solution:** specific constants fixed by initial/boundary conditions
- **Singular solution:** cannot be obtained from the general solution for any value of constants
- Order of ODE formed from $y = Ae^{2x} + Be^{-3x}$: **order 2** (two constants A, B)

1.1. MCQs

Q1. The order and degree of the ODE $\left(\frac{d^2y}{dx^2}\right)^3 + \left(\frac{dy}{dx}\right)^2 = \sin x$ are:

(A) Order 3, Degree 2

(B) Order 2, Degree 3

(C) Order 2, Degree 2

(D) Order 3, Degree 3

Correct Answer: (B) Order 2, Degree 3

Order = highest derivative present = $\frac{d^2y}{dx^2} \Rightarrow$ **order = 2**.

Degree = power of the highest-order derivative = $\left(\frac{d^2y}{dx^2}\right)^{\mathbf{3}} \Rightarrow$ **degree = 3**.

Q2. The ODE formed by eliminating the constant A from $y = Ae^{3x}$ is:

(A) $\frac{dy}{dx} = 3y$

(B) $\frac{dy}{dx} = 3x$

(C) $\frac{dy}{dx} + 3y = 0$

(D) $\dfrac{dy}{dx} = Ae^{3x}$

Correct Answer: (A) $\dfrac{dy}{dx} = 3y$

Differentiate: $y' = 3Ae^{3x} = 3y$ (since $Ae^{3x} = y$).
Eliminating A: $\boxed{y' = 3y}$. This is a first-order ODE (one constant $\Rightarrow$ order 1).

Q3. The general solution of an ODE of order 3 contains:

(A) One arbitrary constant

(B) Two arbitrary constants

(C) Three arbitrary constants

(D) No arbitrary constants

Correct Answer: (C) Three arbitrary constants

The general solution of an nth-order ODE contains exactly n independent arbitrary constants. Order 3 $\Rightarrow$ 3 constants. Each integration introduces one constant.

Q4. The degree of the ODE $\sqrt{1 + (y')^2} = y''$ is:

(A) 1

(B) 2

(C) 3

(D) Not defined

Correct Answer: (B) 2

Square both sides to remove the radical: $1 + (y')^2 = (y'')^2$.
Now the highest-order term is $(y'')^2$, so **degree = 2**. Always clear radicals before determining degree — degree is defined only for polynomial-type ODEs.

Quick Check: Order = look at the highest subscript/prime count. Degree = power of that highest-order term *after* clearing all radicals and fractions. Degree is undefined if the ODE cannot be made polynomial in derivatives (e.g., $e^{y''} = x$).

2. Separable Equations

An ODE is **separable** if it can be written as:

$$g(y)\ \mathrm{d}y = f(x)\ \mathrm{d}x.$$

Method: Separate variables so all y-terms are on one side and all x-terms on the other, then integrate both sides. Add a single constant C (not $+C$ on both sides).

Key Formulas & Results

Standard Procedure:

$$\frac{dy}{dx} = f(x)\cdot g(y) \implies \int \frac{dy}{g(y)} = \int f(x)\,dx + C$$

Useful Integrals:

- $\int \frac{dy}{y} = \ln|y| + C$
- $\int e^{ay}\,dy = \frac{e^{ay}}{a} + C$
- $\int \frac{dy}{a^2+y^2} = \frac{1}{a}\tan^{-1}\frac{y}{a} + C$
- $\int \frac{dy}{y^2-a^2} = \frac{1}{2a}\ln\left|\frac{y-a}{y+a}\right| + C$

2.1. MCQs

Q5. The general solution of $\frac{dy}{dx} = \frac{y}{x}$ is:

(A) $y = x + C$

(B) $y = Cx$

(C) $y = Ce^x$

(D) $y^2 = x^2 + C$

Correct Answer: (B) $y = Cx$

Separate: $\frac{dy}{y} = \frac{dx}{x}$. Integrate: $\ln|y| = \ln|x| + \ln|C| = \ln|Cx|$.
Therefore $y = Cx$ (absorbing sign into C).

Q6. Solve $\frac{dy}{dx} = e^{x+y}$. The general solution is:

(A) $e^{-y} + e^{x} = C$

(B) $e^{-y} = e^{x} + C$

(C) $e^{y} = e^{x} + C$

(D) $y = e^{x} + C$

Correct Answer: (B) $e^{-y} = e^{x} + C$

Write $e^{x+y} = e^{x} \cdot e^{y}$. Separate: $e^{-y}\,\mathrm{d}y = e^{x}\,\mathrm{d}x$.
Integrate: $-e^{-y} = e^{x} + K \;\Rightarrow\; e^{-y} = -e^{x} - K$. Setting $C = -K$: $\boxed{e^{-y} = e^{x} + C}$.
(Exact form depends on sign convention for C.)

Q7. The solution of $\dfrac{dy}{dx} = xy$ with $y(0) = 2$ is:

(A) $y = 2e^{x^2}$

(B) $y = 2e^{x^2/2}$

(C) $y = e^{x^2/2} + 1$

(D) $y = 2x^2$

Correct Answer: (B) $y = 2e^{x^2/2}$

Separate: $\dfrac{dy}{y} = x\,dx$. Integrate: $\ln y = \dfrac{x^2}{2} + C_1 \Rightarrow y = Ce^{x^2/2}$.
IC $y(0) = 2$: $2 = Ce^{0} = C$. So $y = 2e^{x^2/2}$.

Q8. Which ODE is separable?

(A) $\dfrac{dy}{dx} = x + y$

(B) $\dfrac{dy}{dx} = x^2y^2$

(C) $\dfrac{dy}{dx} = \sin(x + y)$

(D) $\dfrac{dy}{dx} = x^2 + y$

Correct Answer: (B) $\dfrac{dy}{dx} = x^2y^2$

(B) factors as $f(x) \cdot g(y) = x^2 \cdot y^2$ — clearly separable: $\dfrac{dy}{y^2} = x^2\,dx$.
(A): $x + y$ does not factor as a product. (C): $\sin(x + y)$ cannot be separated. (D): $x^2 + y$ does not factor.

3. Homogeneous Equations

The ODE $\dfrac{dy}{dx} = f\left(\dfrac{y}{x}\right)$ is called **homogeneous**. Equivalently, $M(x,y)\,dx + N(x,y)\,dy = 0$ is homogeneous if M and N are homogeneous functions of the same degree.

Method: Substitute $y = vx$ (so $\dfrac{dy}{dx} = v + x\dfrac{dv}{dx}$) to convert to a separable ODE in v and x.

Key Formulas & Results

- Substitution: $v = \dfrac{y}{x}, \quad y = vx, \quad \dfrac{dy}{dx} = v + x\dfrac{dv}{dx}$
- After substitution: $v + x\dfrac{dv}{dx} = f(v) \Rightarrow \dfrac{dv}{f(v) - v} = \dfrac{dx}{x}$
- Separate and integrate, then back-substitute $v = y/x$
- A function $f(x,y)$ is **homogeneous of degree** n if $f(tx,ty) = t^n f(x,y)$

3.1. MCQs

Q9. The substitution used to solve a homogeneous ODE $\dfrac{dy}{dx} = f(y/x)$ is:

(A) $y = vx^2$

(B) $y = v + x$

(C) $y = vx$

(D) $x = vy$

Correct Answer: (C) $y = vx$

The standard substitution for homogeneous ODEs is $y = vx$, giving $\dfrac{dy}{dx} = v + x\dfrac{dv}{dx}$. Substituting converts the equation to separable form in v and x.

Q10. The ODE $\dfrac{dy}{dx} = \dfrac{x^2 + y^2}{2xy}$ is:

(A) Linear

(B) Exact

(C) Homogeneous

(D) Bernoulli

Correct Answer: (C) Homogeneous

Divide numerator and denominator by x^2:

$$\frac{dy}{dx} = \frac{1 + (y/x)^2}{2(y/x)} = f\left(\frac{y}{x}\right).$$

Since the RHS depends only on y/x, the equation is **homogeneous**. Substitute $y = vx$ to solve.

Q11. After substituting $y = vx$ in a homogeneous ODE, the resulting equation is always:

(A) Linear in v

(B) Separable in v and x

(C) Exact in v and x

(D) Bernoulli in v

Correct Answer: (B) Separable in v and x

After substitution, the ODE becomes $x\frac{dv}{dx} = f(v) - v$, which separates as: $\frac{dv}{f(v) - v} = \frac{dx}{x}$. Both sides can be integrated independently.

Q12. The function $f(x, y) = x^2 + xy + y^2$ is homogeneous of degree:

(A) 1

(B) 2

(C) 3

(D) 0

Correct Answer: (B) 2

$f(tx, ty) = (tx)^2 + (tx)(ty) + (ty)^2 = t^2x^2 + t^2xy + t^2y^2 = t^2(x^2 + xy + y^2) = t^2 f(x, y)$. So f is homogeneous of degree **2**.

4. Exact Equations & Integrating Factors

Core Concepts

The ODE $M(x, y)\,\mathrm{d}x + N(x, y)\,\mathrm{d}y = 0$ is **exact** if $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$. Then a function $F(x, y)$ exists such that $\frac{\partial F}{\partial x} = M$ and $\frac{\partial F}{\partial y} = N$, and the general solution is

$F(x, y) = C$.

If not exact, an **integrating factor** μ is found such that $\mu M\, dx + \mu N\, dy = 0$ becomes exact.

Key Formulas & Results

Exactness Test: $\dfrac{\partial M}{\partial y} = \dfrac{\partial N}{\partial x}$

Finding F: $F(x, y) = \int M\, dx + g(y)$, then find $g(y)$ from $\dfrac{\partial F}{\partial y} = N$.

Integrating Factors:

- If $\dfrac{M_y - N_x}{N} = h(x)$ only, then $\mu = e^{\int h(x)\, dx}$
- If $\dfrac{N_x - M_y}{M} = k(y)$ only, then $\mu = e^{\int k(y)\, dy}$
- Special cases: $\mu = x^n$, $\mu = y^n$, $\mu = x^m y^n$

4.1. MCQs

Q13. The ODE $(2xy + y^2)\, dx + (x^2 + 2xy)\, dy = 0$ is exact because:

(A) $M = N$

(B) $\dfrac{\partial M}{\partial y} = \dfrac{\partial N}{\partial x} = 2x + 2y$

(C) $\dfrac{\partial M}{\partial x} = \dfrac{\partial N}{\partial y}$

(D) Both M and N are linear

Correct Answer: (B)

$M = 2xy + y^2$: $\dfrac{\partial M}{\partial y} = 2x + 2y$.

$N = x^2 + 2xy$: $\dfrac{\partial N}{\partial x} = 2x + 2y$.

Since $\dfrac{\partial M}{\partial y} = \dfrac{\partial N}{\partial x} = 2x + 2y$, the equation is exact.

Q14. For the exact ODE $(3x^2 + y)\, dx + (x)\, dy = 0$, the solution $F(x, y) = C$ is:

(A) $x^3 + xy = C$

(B) $3x^2 y + x = C$

(C) $x^3 y + x = C$

(D) $x^3 + y = C$

Correct Answer: (A) $x^3 + xy = C$

$M = 3x^2 + y,\ N = x$. Check: $M_y = 1 = N_x$ (exact).
$F = \int M\,dx = \int (3x^2 + y)\,dx = x^3 + xy + g(y)$.
$\dfrac{\partial F}{\partial y} = x + g'(y) = N = x \Rightarrow g'(y) = 0 \Rightarrow g(y) = \text{const}$.
Solution: $\boxed{x^3 + xy = C}$.

Q15. The ODE $(y)\,\mathrm{d}x + (2x - ye^y)\,\mathrm{d}y = 0$ is not exact. The integrating factor $\mu(x)$ is found when $\dfrac{M_y - N_x}{N}$ depends only on x. Here $M_y - N_x =$:

(A) 2

(B) -1

(C) 1

(D) y

Correct Answer: (B) -1

$M = y,\ N = 2x - ye^y$.
$\dfrac{\partial M}{\partial y} = 1$; $\dfrac{\partial N}{\partial x} = 2$.
$M_y - N_x = 1 - 2 = -1$.
$\dfrac{M_y - N_x}{N} = \dfrac{-1}{2x - ye^y}$ — this depends on both x and y, so an x-only integrating factor does not immediately apply. The value of $M_y - N_x$ is -1.

Q16. Which condition makes $M\,dx + N\,dy = 0$ exact?

(A) $M_x = N_y$

(B) $M_y = N_x$

(C) $M = N$

(D) $M_y + N_x = 0$

Correct Answer: (B) $M_y = N_x$

By Clairaut's theorem, if $F_{xy} = F_{yx}$, then $M\,\mathrm{d}x + N\,\mathrm{d}y$ is an exact differential dF iff $\dfrac{\partial M}{\partial y} = \dfrac{\partial N}{\partial x}$. This is the fundamental exactness criterion.

FPSC Shortcut for Exactness: Always compute $\dfrac{\partial M}{\partial y}$ and $\dfrac{\partial N}{\partial x}$ first. If they match, proceed with integration; if not, compute $\dfrac{M_y - N_x}{N}$—if x-only, use integrating factor $\mu = e^{\int h(x)\,dx}$.

5. Linear First-Order ODEs & Bernoulli Equation

A **linear first-order ODE** has the standard form:

$$\frac{dy}{dx} + P(x)\,y = Q(x).$$

Method: Multiply by integrating factor $\mu = e^{\int P\,dx}$; the LHS becomes $\dfrac{d}{dx}(\mu y)$. The **Bernoulli equation** $y' + P(x)y = Q(x)y^n$ ($n \neq 0, 1$) is reduced to linear form by the substitution $v = y^{1-n}$.

Key Formulas & Results

Linear ODE Solution:

$$\mu = e^{\int P(x)\,dx}, \qquad y = \frac{1}{\mu}\left[\int \mu\,Q(x)\,dx + C\right]$$

Bernoulli Equation $y' + Py = Qy^n$:

- Divide by y^n: $y^{-n}y' + Py^{1-n} = Q$
- Let $v = y^{1-n}$: $\dfrac{dv}{dx} = (1-n)y^{-n}\dfrac{dy}{dx}$
- Transformed linear ODE: $\dfrac{dv}{dx} + (1-n)Pv = (1-n)Q$

5.1. MCQs

Q17. The integrating factor for $\dfrac{dy}{dx} + \dfrac{3}{x}y = x^2$ is:

(A) e^{3x}

(B) x^3

(C) $e^{3/x}$

(D) x^{-3}

Correct Answer: (B) x^3

$P(x) = \dfrac{3}{x}$. Integrating factor: $\mu = e^{\int \frac{3}{x}\,dx} = e^{3\ln x} = e^{\ln x^3} = x^3$.

Q18. The general solution of $\dfrac{dy}{dx} - 2y = e^{2x}$ is:

(A) $y = e^{2x}(x + C)$

(B) $y = xe^{2x} + Ce^{2x}$

(C) $y = (x + C)e^{2x}$

(D) Both (A) and (C) are the same

Correct Answer: (D) Both (A) and (C) are the same, i.e., $y = (x + C)e^{2x}$

$\mu = e^{\int -2\,dx} = e^{-2x}$. Multiply: $\dfrac{d}{dx}(e^{-2x}y) = e^{-2x} \cdot e^{2x} = 1$.
Integrate: $e^{-2x}y = x + C \Rightarrow y = (x + C)e^{2x}$.

Q19. The Bernoulli equation $\dfrac{dy}{dx} + y = y^3$ is reduced to a linear ODE by the substitution:

(A) $v = y^2$

(B) $v = y^{-2}$

(C) $v = y^3$

(D) $v = 1/y$

Correct Answer: (B) $v = y^{-2}$

Here $n = 3$. Use $v = y^{1-n} = y^{1-3} = y^{-2}$.
Then $\dfrac{dv}{dx} = -2y^{-3}\dfrac{dy}{dx}$. Dividing the Bernoulli ODE by y^3 and substituting $v = y^{-2}$ gives the linear ODE $\dfrac{dv}{dx} - 2v = -2$.

Q20. The ODE $\dfrac{dy}{dx} + y\cos x = \frac{1}{2}\sin 2x$ is:

(A) Separable

(B) Bernoulli

(C) Linear

(D) Homogeneous

Correct Answer: (C) Linear

The ODE has the form $y' + P(x)y = Q(x)$ with $P(x) = \cos x$ and $Q(x) = \frac{1}{2}\sin 2x = \sin x \cos x$. This is a **linear** first-order ODE. Integrating factor: $\mu = e^{\int \cos x\,dx} = e^{\sin x}$.

6. Orthogonal Trajectories

Two families of curves are **orthogonal trajectories** of each other if every curve in one family intersects every curve in the other family at right angles. At the intersection point, the slopes are negative reciprocals: $m_1 \cdot m_2 = -1$.

Method: (1) Find ODE of given family $y' = f(x, y)$. (2) Replace y' by $-\frac{1}{y'}$ to get ODE of orthogonal family. (3) Solve the new ODE.

Key Formulas & Results

- Given family ODE: $\frac{dy}{dx} = f(x, y)$
- Orthogonal trajectory ODE: $\frac{dy}{dx} = -\frac{1}{f(x, y)}$
- Orthogonal trajectories of circles through origin ($y = mx$): are circles centred on origin
- **Classic pair:** Parabolas $y = cx^2$ and ellipses $x^2 + \frac{y^2}{2} = k$

6.1. MCQs

Q21. The orthogonal trajectories of the family $y = cx^2$ satisfy the ODE:

(A) $\frac{dy}{dx} = \frac{2y}{x}$

(B) $\frac{dy}{dx} = -\frac{x}{2y}$

(C) $\frac{dy}{dx} = \frac{x}{2y}$

(D) $\frac{dy}{dx} = cx$

Correct Answer: (B) $\frac{dy}{dx} = -\frac{x}{2y}$

Given family: $y = cx^2$. Eliminate c: $c = \frac{y}{x^2}$, so $y' = 2cx = \frac{2y}{x}$.

Orthogonal trajectory: replace y' by $-\frac{1}{y'}$: $\frac{dy}{dx} = -\frac{x}{2y}$.

Q22. The orthogonal trajectories of the family of circles $x^2 + y^2 = c^2$ are:

(A) Ellipses

(B) Parabolas

(C) Straight lines through the origin

(D) Hyperbolas

Correct Answer: (C) Straight lines through the origin

$x^2 + y^2 = c^2$: differentiate $\Rightarrow 2x + 2yy' = 0 \Rightarrow y' = -\dfrac{x}{y}$.

Orthogonal ODE: $y' = \dfrac{y}{x} \Rightarrow \dfrac{dy}{y} = \dfrac{dx}{x} \Rightarrow y = Kx$. These are straight lines through the origin.

Q23. For two curves to be orthogonal at a point, their slopes m_1 and m_2 satisfy:

(A) $m_1 = m_2$

(B) $m_1 + m_2 = 0$

(C) $m_1 \cdot m_2 = -1$

(D) $m_1 \cdot m_2 = 1$

Correct Answer: (C) $m_1 \cdot m_2 = -1$

Two lines are perpendicular iff their slopes are **negative reciprocals**: $m_2 = -\dfrac{1}{m_1}$, i.e., $m_1 \cdot m_2 = -1$. This is the geometric foundation of orthogonal trajectories.

7. Second-Order Linear ODEs: Homogeneous with Constant Coefficients

The ODE $ay''+by'+cy = 0$ (a, b, c constants, $a \neq 0$) is solved via the **characteristic (auxiliary) equation** $ar^2 + br + c = 0$. The nature of roots r_1, r_2 determines the form of the general solution.

Key Formulas & Results

Let $D = b^2 - 4ac$ be the discriminant:

Case	Roots	General Solution
$D > 0$ (**over-damped**)	$r_1 \neq r_2$ (real)	$C_1e^{r_1x} + C_2e^{r_2x}$
$D = 0$ (**critically damped**)	$r_1 = r_2 = r$ (repeated)	$(C_1 + C_2x)e^{rx}$
$D < 0$ (**under-damped**)	$\alpha \pm \beta i$ (complex)	$e^{\alpha x}(C_1 \cos \beta x + C_2 \sin \beta x)$

7.1. MCQs

Q24. The general solution of $y'' - 4y' + 4y = 0$ is:

(A) $C_1e^{2x} + C_2e^{-2x}$

(B) $(C_1 + C_2x)e^{2x}$

(C) $e^{2x}(C_1 \cos 2x + C_2 \sin 2x)$

(D) $C_1 \cos 2x + C_2 \sin 2x$

Correct Answer: (B) $(C_1 + C_2x)e^{2x}$

Characteristic equation: $r^2 - 4r + 4 = (r - 2)^2 = 0 \Rightarrow r = 2$ (repeated).

Repeated root $r = 2$: general solution is $\boxed{(C_1 + C_2x)e^{2x}}$ — critically damped.

Q25. The general solution of $y'' + 2y' + 5y = 0$ is:

(A) $e^{-x}(C_1 \cos 2x + C_2 \sin 2x)$

(B) $C_1e^{-x} + C_2e^{-5x}$

(C) $(C_1 + C_2x)e^{-x}$

(D) $e^{x}(C_1 \cos 2x + C_2 \sin 2x)$

Correct Answer: (A) $e^{-x}(C_1 \cos 2x + C_2 \sin 2x)$

Characteristic: $r^2 + 2r + 5 = 0 \Rightarrow r = \dfrac{-2 \pm \sqrt{4 - 20}}{2} = \dfrac{-2 \pm \sqrt{-16}}{2} = -1 \pm 2i$.

Complex roots $\alpha = -1$, $\beta = 2$: solution $e^{-x}(C_1 \cos 2x + C_2 \sin 2x)$.

Q26. For the ODE $y'' + 9y = 0$, the general solution is:

(A) $C_1e^{3x} + C_2e^{-3x}$

(B) $(C_1 + C_2x)e^{3x}$

(C) $C_1 \cos 3x + C_2 \sin 3x$

(D) $C_1e^{3x} + C_2xe^{3x}$

Correct Answer: (C) $C_1 \cos 3x + C_2 \sin 3x$

Characteristic: $r^2 + 9 = 0 \Rightarrow r = \pm 3i$ (pure imaginary, $\alpha = 0, \beta = 3$).
Solution: $e^{0 \cdot x}(C_1 \cos 3x + C_2 \sin 3x) = \boxed{C_1 \cos 3x + C_2 \sin 3x}$.

Q27. The characteristic equation of $2y'' - 3y' - 2y = 0$ has roots:

(A) $r = 2$ and $r = -\frac{1}{2}$

(B) $r = -2$ and $r = \frac{1}{2}$

(C) $r = 1$ and $r = -1$

(D) $r = 3$ and $r = -2$

Correct Answer: (A) $r = 2$ **and** $r = -\frac{1}{2}$

$2r^2 - 3r - 2 = 0$. Factor: $(2r + 1)(r - 2) = 0 \Rightarrow r = 2$ or $r = -\dfrac{1}{2}$.
General solution: $y = C_1 e^{2x} + C_2 e^{-x/2}$.

Discriminant Trick: $b^2 - 4ac$: positive $\to$ two real roots; zero $\to$ repeated root; negative $\to$ complex conjugate roots. For $y'' + \omega^2 y = 0$ (pure oscillation), roots are always $\pm \omega i$.

8. Method of Undetermined Coefficients

For $ay'' + by' + cy = f(x)$ where $f(x)$ is a polynomial, exponential, sine, cosine, or their products. Assume a particular solution y_p of the same "type" as $f(x)$, substitute into the ODE, and match coefficients. **Modification Rule:** If y_p overlaps with y_h, multiply y_p by x (or x^2 if the overlap is a repeated term).

Key Formulas & Results

$f(x)$	Assumed y_p
$P_n(x)$ **(degree** n**)**	$A_nx^n + \cdots + A_1x + A_0$
e^{ax}	Ae^{ax}
$\sin bx$ **or** $\cos bx$	$A\sin bx + B\cos bx$
$e^{ax}\sin bx$ **or** $e^{ax}\cos bx$	$e^{ax}(A\cos bx + B\sin bx)$
x^ne^{ax}	$(A_nx^n + \cdots + A_0)e^{ax}$

8.1. MCQs

Q28. The particular solution form assumed for $y'' - 3y' + 2y = 4e^{3x}$ is:

(A) Ae^{3x}

(B) Axe^{3x}

(C) $Ae^{3x} + Be^{x}$

(D) $(Ax + B)e^{3x}$

Correct Answer: (A) Ae^{3x}

Homogeneous solution: $r^2 - 3r + 2 = (r-1)(r-2) = 0 \Rightarrow y_h = C_1e^x + C_2e^{2x}$.
$f(x) = 4e^{3x}$: Since $r = 3$ is **not** a root of the characteristic equation, assume $y_p = Ae^{3x}$ without modification.
Substituting: $9Ae^{3x} - 9Ae^{3x} + 2Ae^{3x} = 4e^{3x} \Rightarrow 2A = 4 \Rightarrow A = 2$.

Q29. For $y'' - 4y = \sin 2x$, the assumed form of y_p is:

(A) $A\sin 2x$

(B) $A\cos 2x + B\sin 2x$

(C) $Ax\sin 2x + Bx\cos 2x$

(D) $A\sin 2x + B\cos 2x + Ce^{2x}$

Correct Answer: (B) $A\cos 2x + B\sin 2x$

$f(x) = \sin 2x$ — always assume *both* sin and cos with the same argument.
Char. roots: $r^2 - 4 = 0 \Rightarrow r = \pm 2$ (real). Since $\pm 2i$ are not roots, no modification is needed. Assume $y_p = A\cos 2x + B\sin 2x$.

Q30. If $f(x) = xe^x$ and e^x appears in y_h, the correct form of y_p is:

(A) Axe^x

(B) $(Ax + B)e^x$

(C) $x(Ax + B)e^x$

(D) $(Ax^2 + Bx)e^x$

Correct Answer: (D) $(Ax^2 + Bx)e^x$

For $f(x) = xe^x$, the "natural" guess is $(Ax+B)e^x$. But if e^x (and hence xe^x) is part of y_h (e.g., repeated root $r = 1$), multiply by x: $y_p = x(Ax+B)e^x = (Ax^2+Bx)e^x$. This is the **Modification Rule**.

Q31. For $y'' + y = \cos x$ (resonance case), the correct y_p form is:

(A) $A\cos x + B\sin x$

(B) $Ax\cos x$

(C) $Ax\sin x + Bx\cos x$

(D) $Ax\cos x + Bx\sin x$

Correct Answer: (D) $Ax\cos x + Bx\sin x$

Char. equation $r^2 + 1 = 0 \Rightarrow r = \pm i$. The natural guess $A\cos x + B\sin x$ overlaps with y_h. Apply Modification Rule: multiply by x, giving $y_p = x(A\cos x + B\sin x) = Ax\cos x + Bx\sin x$. This resonance case produces growing oscillations.

9. Variation of Parameters

For $y'' + P(x)y' + Q(x)y = f(x)$ with homogeneous solution $y_h = C_1y_1 + C_2y_2$. Assume $y_p = u_1(x)y_1 + u_2(x)y_2$ where:

$$u_1' = -\frac{y_2 f}{W}, \qquad u_2' = \frac{y_1 f}{W}, \qquad W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = y_1y_2' - y_2y_1'.$$

Then $u_1 = \int u_1'\,dx$, $u_2 = \int u_2'\,dx$, and $y_p = u_1y_1 + u_2y_2$.

Key Formulas & Results

- Works for *any* continuous $f(x)$ — more general than undetermined coefficients
- $W[y_1, y_2] \neq 0$ guaranteed if y_1, y_2 are linearly independent (Abel's theorem)
- $W' + (P)W = 0 \Rightarrow W = Ce^{-\int P\,dx}$ (Abel's Identity)
- Standard form required: coefficient of y'' must be **1** before applying the formula

9.1. MCQs

Q32. In variation of parameters for $y'' + y = \sec x$, the Wronskian $W[y_1, y_2]$ with $y_1 = \cos x$, $y_2 = \sin x$ is:

(A) 0

(B) -1

(C) 1

(D) $\cos^2 x - \sin^2 x$

Correct Answer: (C) 1

$W = y_1 y_2' - y_2 y_1' = \cos x \cdot \cos x - \sin x \cdot (-\sin x) = \cos^2 x + \sin^2 x = 1.$

Q33. For $y'' + y = \tan x$, using variation of parameters with $y_h = C_1 \cos x + C_2 \sin x$, the formula for u_2' is:

(A) $u_2' = \cos x \tan x$

(B) $u_2' = \sin x \tan x$

(C) $u_2' = \cos x$

(D) $u_2' = -\sin x \tan x$

Correct Answer: (A) $u_2' = \cos x \tan x = \sin x$

$y_1 = \cos x$, $y_2 = \sin x$, $f(x) = \tan x$, $W = 1$.

$u_2' = \dfrac{y_1 f}{W} = \dfrac{\cos x \cdot \tan x}{1} = \cos x \cdot \dfrac{\sin x}{\cos x} = \sin x.$

So option (A) correctly states $u_2' = \cos x \tan x$ (which simplifies to $\sin x$).

Q34. Variation of parameters is preferred over undetermined coefficients when $f(x)$ is:

(A) A polynomial

(B) An exponential

(C) $\sec x$ or $\ln x$

(D) A constant

Correct Answer: (C) $\sec x$ **or** $\ln x$

Undetermined coefficients works only when $f(x)$ belongs to specific families (polynomials, exponentials, sines/cosines). Functions like $\sec x$, $\tan x$, $\ln x$, $\dfrac{1}{x}$ require **variation of parameters**, which works for any continuous $f(x)$.

10. Cauchy–Euler Equation

The **Cauchy–Euler** (equidimensional) equation has the form:

$$ax^2y'' + bxy' + cy = f(x).$$

Method: Substitute $x = e^t$ (or $y = x^m$) to transform to an ODE with constant coefficients. With $y = x^m$: substitute into the homogeneous equation and solve the **indicial equation** $am(m-1) + bm + c = 0$.

Key Formulas & Results

Let $t = \ln x$ (so $x = e^t$), then:

$$x\frac{dy}{dx} = \frac{dy}{dt}, \qquad x^2\frac{d^2y}{dx^2} = \frac{d^2y}{dt^2} - \frac{dy}{dt}.$$

Indicial equation $am(m-1) + bm + c = 0$ (from substituting $y = x^m$):

- Two distinct real roots $m_1 \neq m_2$: $y = C_1x^{m_1} + C_2x^{m_2}$
- Repeated root $m_1 = m_2 = m$: $y = (C_1 + C_2 \ln x)x^m$
- Complex roots $m = \alpha \pm \beta i$: $y = x^\alpha[C_1 \cos(\beta \ln x) + C_2 \sin(\beta \ln x)]$

10.1. MCQs

Q35. The general solution of $x^2y'' - 3xy' + 4y = 0$ is:

(A) $C_1x^2 + C_2x^2 \ln x$

(B) $C_1x + C_2x^4$

(C) $(C_1 + C_2 \ln x)x^2$

(D) $C_1x^{-2} + C_2x^2$

Correct Answer: (C) $(C_1 + C_2 \ln x)x^2$

Indicial equation: $m(m-1) - 3m + 4 = 0 \Rightarrow m^2 - 4m + 4 = (m-2)^2 = 0 \Rightarrow m = 2$ (repeated).

Repeated root $m = 2$: $y = (C_1 + C_2 \ln x)x^2$.

Q36. The indicial equation for $x^2y'' + xy' + y = 0$ is:

(A) $m^2 + 1 = 0$

(B) $m^2 - m + 1 = 0$

(C) $m^2 + 0 \cdot m + 1 = 0$

(D) $m(m-1) + m + 1 = 0$

Correct Answer: (D) $m(m-1)+m+1=0$**, i.e.,** $m^2+1=0$

$a=1, b=1, c=1$. Indicial equation: $m(m-1)+1\cdot m+1 = m^2-m+m+1 = m^2+1=0$.
Roots: $m=\pm i$ (pure imaginary, $\alpha=0, \beta=1$).
Solution: $x^0[C_1\cos(\ln x)+C_2\sin(\ln x)] = C_1\cos(\ln x)+C_2\sin(\ln x)$.

Q37. The Cauchy–Euler equation $x^2y''+xy'-y=0$ has solution:

(A) $C_1x+C_2x^{-1}$

(B) $(C_1+C_2\ln x)x$

(C) $C_1x+C_2\ln x$

(D) $C_1x^2+C_2x^{-1}$

Correct Answer: (A) $C_1x+C_2x^{-1}$

Indicial: $m(m-1)+m-1=0 \Rightarrow m^2-1=0 \Rightarrow m=\pm 1$.

Two distinct real roots: $y = C_1x^1+C_2x^{-1} = C_1x+\dfrac{C_2}{x}$.

Recognizing Cauchy–Euler: The key feature is that each term has $x^ny^{(n)}$, i.e., the power of x matches the order of the derivative. Always form the indicial equation $m(m-1)+bm+c=0$ first.

11. Power Series Solutions

For ODEs with variable coefficients near an **ordinary point** $x=0$: assume $y = \sum_{n=0}^{\infty} a_nx^n$, substitute into ODE, collect like powers of x, and find the **recurrence relation** for coefficients a_n.

A point $x=x_0$ is **ordinary** if $P(x)$ and $Q(x)$ in $y''+Py'+Qy=0$ are analytic there. If $x(x-x_0)P$ and x^2Q are analytic, then x_0 is a **regular singular point** (Frobenius method applies).

Key Formulas & Results

- Assume: $y=\sum_{n=0}^{\infty} a_nx^n$, then $y'=\sum_{n=1}^{\infty} na_nx^{n-1}$, $y''=\sum_{n=2}^{\infty} n(n-1)a_nx^{n-2}$
- Shift index to align powers: $\sum_{n=2}^{\infty} n(n-1)a_nx^{n-2} = \sum_{n=0}^{\infty}(n+2)(n+1)a_{n+2}x^n$

- **Frobenius method:** $y = x^r \sum_{n=0}^{\infty} a_n x^n$; indicial equation gives r
- **Convergence:** radius of convergence $\geq$ distance from x_0 to nearest singular point

11.1. MCQs

Q38. For the power series solution $y = \sum_{n=0}^{\infty} a_n x^n$ of $y'' + y = 0$, the recurrence relation is:

(A) $a_{n+2} = \dfrac{a_n}{n+1}$

(B) $a_{n+2} = -\dfrac{a_n}{(n+2)(n+1)}$

(C) $a_{n+1} = -\dfrac{a_n}{n+1}$

(D) $a_{n+2} = \dfrac{a_n}{(n+2)!}$

Correct Answer: (B) $a_{n+2} = -\dfrac{a_n}{(n+2)(n+1)}$

Substituting $y = \sum a_n x^n$ into $y'' + y = 0$: $\sum (n+2)(n+1)a_{n+2}x^n + \sum a_n x^n = 0$.
Equating: $(n+2)(n+1)a_{n+2} + a_n = 0 \Rightarrow a_{n+2} = -\dfrac{a_n}{(n+2)(n+1)}$.
This generates $\cos x$ and $\sin x$ series (as expected for $y'' + y = 0$).

Q39. The point $x = 0$ is a regular singular point of $x^2y'' + xy' + (x^2 - n^2)y = 0$ (Bessel's equation) because:

(A) $P(x)$ and $Q(x)$ are polynomials

(B) $xP(x) = 1$ and $x^2Q(x) = x^2 - n^2$ are analytic at $x = 0$

(C) The equation has constant coefficients

(D) The indicial equation has integer roots

Correct Answer: (B)

Standard form: $y'' + \dfrac{1}{x}y' + \dfrac{x^2 - n^2}{x^2}y = 0$. Here $P = \dfrac{1}{x}$ has a singularity at $x = 0$. But $xP(x) = 1$ and $x^2Q(x) = x^2 - n^2$ are both analytic at $x = 0$, so $x = 0$ is a **regular singular point** — Frobenius method applies.

Q40. The power series solution of $y' - y = 0$ about $x = 0$ gives:

(A) $y = a_0(1 + x + x^2 + x^3 + \cdots) = a_0 e^x$

(B) $y = a_0 \cos x$

(C) $y = a_0 \ln x$

(D) $y = a_0(1 - x^2/2 + \cdots)$

Correct Answer: (A) $y = a_0 e^x$

From $y' - y = 0$: recurrence $(n+1)a_{n+1} - a_n = 0 \Rightarrow a_{n+1} = \dfrac{a_n}{n+1}$.

So $a_n = \dfrac{a_0}{n!}$ and $y = a_0 \sum_{n=0}^{\infty} \dfrac{x^n}{n!} = a_0 e^x$.

12. Laplace Transforms

The **Laplace transform** of $f(t)$ for $t \geq 0$ is:

$$\mathcal{L}\{f(t)\} = F(s) = \int_0^\infty e^{-st} f(t)\,\mathrm{d}t, \quad s > 0.$$

It converts ODEs with initial conditions into algebraic equations in s, which are easier to solve. The solution in t-domain is recovered via the **inverse Laplace transform**.

Key Formulas & Results

$f(t)$	$F(s) = \mathcal{L}\{f\}$	$f(t)$	$F(s)$
1	$\dfrac{1}{s}$	e^{at}	$\dfrac{1}{s-a}$
t^n	$\dfrac{n!}{s^{n+1}}$	$t^n e^{at}$	$\dfrac{n!}{(s-a)^{n+1}}$
$\sin bt$	$\dfrac{b}{s^2+b^2}$	$e^{at}\sin bt$	$\dfrac{b}{(s-a)^2+b^2}$
$\cos bt$	$\dfrac{s}{s^2+b^2}$	$e^{at}\cos bt$	$\dfrac{s-a}{(s-a)^2+b^2}$

Key Properties: $\mathcal{L}\{f'\} = sF(s) - f(0)$; $\mathcal{L}\{f''\} = s^2F(s) - sf(0) - f'(0)$; $\mathcal{L}\{e^{at}f(t)\} = F(s-a)$

12.1. MCQs

Q41. $\mathcal{L}\{\sin 2t\}$ equals:

(A) $\dfrac{s}{s^2+4}$

(B) $\dfrac{2}{s^2+4}$

(C) $\dfrac{2}{s^2-4}$

(D) $\dfrac{4}{s^2+4}$

Correct Answer: (B) $\dfrac{2}{s^2+4}$

Standard formula: $\mathcal{L}\{\sin bt\} = \dfrac{b}{s^2+b^2}$. With $b=2$: $\mathcal{L}\{\sin 2t\} = \dfrac{2}{s^2+4}$.

Q42. $\mathcal{L}\{e^{3t}\cos 2t\}$ equals:

(A) $\dfrac{s-3}{(s-3)^2+4}$

(B) $\dfrac{s+3}{(s+3)^2+4}$

(C) $\dfrac{s-3}{(s-3)^2+2}$

(D) $\dfrac{2}{(s-3)^2+4}$

Correct Answer: (A) $\dfrac{s-3}{(s-3)^2+4}$

First shifting theorem: $\mathcal{L}\{e^{at}f(t)\} = F(s-a)$.

$\mathcal{L}\{\cos 2t\} = \dfrac{s}{s^2+4}$. With $a=3$: replace s by $s-3$: $\mathcal{L}\{e^{3t}\cos 2t\} = \dfrac{s-3}{(s-3)^2+4}$.

Q43. If $\mathcal{L}\{y''\} = s^2Y - sy(0) - y'(0)$, and $y(0)=1$, $y'(0)=0$, then $\mathcal{L}\{y''+4y=0\}$ becomes:

(A) $(s^2+4)Y = s$

(B) $(s^2+4)Y = s+1$

(C) $(s^2-4)Y = s$

(D) $s^2Y + 4Y = 0$

Correct Answer: (A) $(s^2+4)Y = s$

$\mathcal{L}\{y''\} + 4\mathcal{L}\{y\} = 0 \Rightarrow [s^2Y - s\cdot 1 - 0] + 4Y = 0$.

$(s^2+4)Y = s \Rightarrow Y = \dfrac{s}{s^2+4} \Rightarrow y = \mathcal{L}^{-1}\left\{\dfrac{s}{s^2+4}\right\} = \cos 2t$.

Q44. $\mathcal{L}^{-1}\left\{\dfrac{1}{s^2+6s+13}\right\}$ equals:

(A) $e^{-3t}\sin 2t$

(B) $\dfrac{1}{2}e^{-3t}\sin 2t$

(C) $e^{3t}\cos 2t$

(D) $e^{-3t}\cos 2t$

Correct Answer: (B) $\frac{1}{2}e^{-3t}\sin 2t$

Complete the square: $s^2 + 6s + 13 = (s+3)^2 + 4 = (s+3)^2 + 2^2$.

$\frac{1}{(s+3)^2+4} = \frac{1}{2} \cdot \frac{2}{(s+3)^2+4}$.

$\mathcal{L}^{-1} = \frac{1}{2}e^{-3t}\sin 2t$.

Q45. The Laplace transform method is especially useful for solving ODEs because:

(A) It only works for constant-coefficient ODEs

(B) It automatically incorporates initial conditions, avoiding the need to find C separately

(C) It requires no inverse transform

(D) It eliminates all constants of integration

Correct Answer: (B)

The key advantage of the Laplace transform is that **initial conditions are built directly** into the transformed equation ($\mathcal{L}\{y'\} = sY - y(0)$, etc.), so the particular solution satisfying the IVP is obtained directly without first finding a general solution and then applying initial conditions separately.

Inverse Transform Trick: Always complete the square for quadratics like $s^2 + bs + c$. Write in the form $(s-\alpha)^2 + \beta^2$ and use the first shifting theorem.

13. Partial Differential Equations (Introduction)

Core Concepts

A **partial differential equation (PDE)** involves an unknown function of *two or more* independent variables and its partial derivatives.

Classic PDEs:

- **Heat equation:** $u_t = k\,u_{xx}$ (parabolic)
- **Wave equation:** $u_{tt} = c^2 u_{xx}$ (hyperbolic)
- **Laplace equation:** $u_{xx} + u_{yy} = 0$ (elliptic)
- **Poisson equation:** $u_{xx} + u_{yy} = f(x,y)$ (elliptic, non-homogeneous)

Order = highest partial derivative. **Linear** if no products of u and its derivatives.

Key Formulas & Results

Method of Separation of Variables: Assume $u(x,t) = X(x) \cdot T(t)$, substitute into PDE, separate: $\dfrac{T'}{kT} = \dfrac{X''}{X} = -\lambda$ (separation constant).

D'Alembert's Solution of $u_{tt} = c^2 u_{xx}$: $u(x,t) = f(x+ct) + g(x-ct)$.

Classification of $Au_{xx} + Bu_{xy} + Cu_{yy} = F$:
$B^2 - 4AC$: < 0 elliptic; $= 0$ parabolic; > 0 hyperbolic.

13.1. MCQs

Q46. The heat equation $u_t = k\, u_{xx}$ is classified as:

(A) Hyperbolic

(B) Elliptic

(C) Parabolic

(D) Ordinary

Correct Answer: (C) Parabolic

For $Au_{xx} + Bu_{xy} + Cu_{yy}$: the heat equation has $A = k$, $B = 0$, $C = 0$ (no u_{yy} term since the other variable is t, not y). The discriminant $B^2 - 4AC = 0 - 0 = 0$, so it is **parabolic**.

Q47. In the method of separation of variables, $u(x,t) = X(x)T(t)$ is substituted into the heat equation. The equation separates into:

(A) $\dfrac{X''}{X} = \dfrac{T'}{kT} = \lambda$

(B) $X'' + T' = 0$

(C) $X''T = XT'$

(D) $\dfrac{X''}{X} = -\lambda$ and $\dfrac{T'}{kT} = -\lambda$ for some constant λ

Correct Answer: (D)

$u_t = kXT'$ and $u_{xx} = X''T$, so $XT' = kX''T \Rightarrow \dfrac{T'}{kT} = \dfrac{X''}{X} = -\lambda$.
Each side depends on a *different* variable, so both must equal the same constant $-\lambda$. The sign is chosen $(-\lambda)$ to ensure bounded/physical solutions.

Q48. The wave equation $u_{tt} = c^2 u_{xx}$ is classified as:

(A) Parabolic

(B) Elliptic

(C) Hyperbolic

(D) Linear and separable only

Correct Answer: (C) Hyperbolic

Wave equation: $A = c^2$, $B = 0$, $C = -1$ (writing as $c^2 u_{xx} - u_{tt} = 0$).
$B^2 - 4AC = 0 - 4(c^2)(-1) = 4c^2 > 0$ — **hyperbolic**.
D'Alembert's general solution: $u(x,t) = f(x+ct) + g(x-ct)$.

Q49. Laplace's equation $\nabla^2 u = u_{xx} + u_{yy} = 0$ describes:

(A) Heat flow in a rod

(B) Vibrating strings

(C) Steady-state (time-independent) temperature distributions

(D) Quantum mechanical wavefunctions

Correct Answer: (C) Steady-state temperature distributions

Laplace's equation arises when the time derivative is zero in the heat equation ($u_t = k\nabla^2 u$ with $u_t = 0$ gives $\nabla^2 u = 0$). It models **steady-state** (equilibrium) phenomena: electrostatics, gravitational potential, and time-independent temperature distributions.

PDE Classification Mnemonic:
Elliptic → **E**quilibrium (Laplace, Poisson) — $B^2 - 4AC < 0$
Parabolic → **P**ropagation of heat (Heat equation) — $B^2 - 4AC = 0$
Hyperbolic → **H**urrying waves (Wave equation) — $B^2 - 4AC > 0$

14. Quick-Reference Master Summary Table

Topic	Key Fact / Formula
Order & Degree	Order = highest derivative; Degree = its power (after clearing radicals)
Separable ODE	$\int g(y)dy = \int f(x)dx + C$
Homogeneous ODE	Substitute $y = vx$; reduces to separable
Exact ODE	$M_y = N_x$; solution $F(x, y) = C$
Integrating Factor	$\mu = e^{\int \frac{M_y - N_x}{N} dx}$ if $h(x)$ only
Linear 1st Order	$\mu = e^{\int P\,dx}$; $\quad y = \frac{1}{\mu}\int \mu Q\,dx + C$
Bernoulli	$v = y^{1-n}$ reduces to linear ODE
Orthogonal Traj.	Replace y' by $-1/y'$ in ODE of given family
Char. Eq. roots	Distinct real: $C_1e^{r_1x} + C_2e^{r_2x}$; Repeated: $(C_1 + C_2x)e^{rx}$; Complex: $e^{\alpha x}(C_1 \cos \beta x + C_2 \sin \beta x)$
Undetermined Coeff.	Assume y_p same type as $f(x)$; multiply by x if overlap with y_h
Var. of Parameters	$u_1' = -y_2 f/W$; $\quad u_2' = y_1 f/W$; $\quad y_p = u_1y_1 + u_2y_2$
Cauchy–Euler	$y = x^m$; indicial eq. $m(m-1) + bm + c = 0$
Power Series	$y = \sum a_n x^n$; find recurrence relation
Laplace Transform	$\mathcal{L}\{f'\} = sF - f(0)$; $\quad \mathcal{L}\{\sin bt\} = b/(s^2 + b^2)$
PDE Classification	$B^2 - 4AC$: < 0 elliptic; $= 0$ parabolic; > 0 hyperbolic

Exam Practice Resource

For further exam practice, the following resource provides a comprehensive collection of solved MCQs:
Access the material here

Best of Luck in Your PSC / FPSC Examination!

Focus on: Characteristic equations, Laplace transforms, Integrating factors, and Exactness.
These four topics account for the majority of ODE questions in PSC Mathematics examinations.